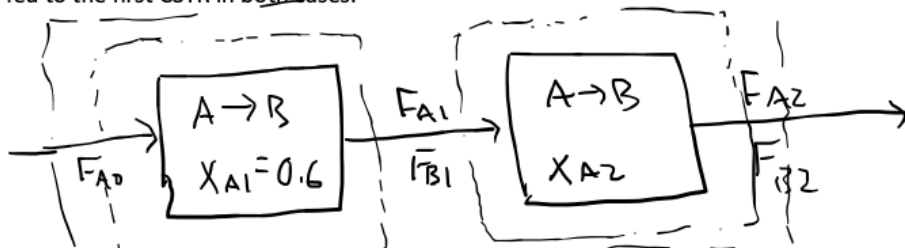


EOX CSTRs in series

The liquid phase reaction $A \rightarrow B$ is carried out isothermally and isobarically in a CSTR to 60% conversion. If a second CSTR is added downstream of the first CSTR, what should the volume of that CSTR be to achieve an overall conversion of 90%? Pure A is fed to the first CSTR in both cases.



$$\text{Want } F_{A2} = F_{A0} (1 - 0.9) = 0.1 F_{A0}$$

$$\text{1st CSTR: } V_1 = \frac{F_{A0} X_{A1}}{-r_A}, \text{ assume elementary kinetics}$$

$$-r_A = k C_{A1} = k \frac{F_{A1}}{V} \quad , \quad F_{A1} = F_{A0} (1 - X_{A1}) = 0.4 F_{A0}$$

$$V_1 = \frac{0.6 F_{A0} V}{k (0.4) F_{A0}} = \frac{2V}{3k}$$

$$\text{2nd CSTR: } V_2 = \frac{F_{A1} X_{A2}}{-r_A}, \quad -r_A = k C_{A2} = k \frac{F_{A2}}{V}$$

$$V_2 = \frac{(0.4) F_{A0} X_{A2} V}{k (0.1) F_{A0}} = \frac{4 X_{A2} V}{k}$$

$$X_{A2} = \frac{F_{A1} - F_{A2}}{F_{A1}} = \frac{0.4 F_{A0} - 0.1 F_{A0}}{0.4 F_{A0}} = 0.75$$

$$V_2 = \frac{3v}{K}$$

$$\frac{V_2}{V_1} = \frac{3v}{K} \cdot \frac{3K}{2v}$$

$$V_2 = 1.5 V_1$$